

CMSC 491/691: From Signals to Insights: Ubiquitous Sensing for Health and Interaction

Syllabus

Spring 2026

Instructor: [Dong Li](#)

Time: Monday & Wednesday 10:00 AM – 11:15 AM

Location: Janet & Walter Sondheim 111

Office Hours: Monday from 2:00 PM to 3:00 PM at ITE 355

Teaching Assistant: TBD

Course Overview

Advances in mobile, wearable, and wireless technologies, together with recent progress in machine learning and foundation models, have transformed sensing into a core capability of ubiquitous computing systems. Signals originally designed for communication or interaction are increasingly repurposed to sense human behavior, physiology, and the physical environment. This course examines how signals are transformed into actionable insights in modern sensing systems.

The course provides a systems-oriented introduction to ubiquitous sensing for health and interaction, covering the full pipeline from physical signal propagation and signal processing to mobile and wearable platforms and AI-enabled inference. Topics include wireless sensing via acoustic, RF, and optical signals; sensing with smartphones, wearables, earables, and smart glasses; and learning-based approaches ranging from machine learning to deep learning, foundation models, and LLM-enabled sensing.

The course is organized into six modules: foundations of signals and sensing; wireless sensing via wave propagation; mobile and wearable sensing platforms; head-worn sensing systems; AI-enabled sensing systems; and emerging paradigms and frontiers in ubiquitous sensing. Students will engage with recent research papers and guest lectures from leading researchers in the field.

A central component of the course is a semester-long, research-oriented project in which students design, implement, and evaluate a sensing system, guided by structured milestones. By the end of the course, students will be equipped to analyze, build, and evaluate real-world sensing systems for health, interaction, and emerging applications.

Communication

I encourage all students to actively participate in classroom discussions, whether by asking questions or sharing your perspectives. Your feedback throughout the semester is valuable to me, and I welcome it as we progress through the course.

- Course announcements, materials, and assignment submissions will be managed through Blackboard.
- For general inquiries or assistance, please use the Blackboard discussion board or email. When emailing, please include the following prefix in the subject line: *[CMSC 491/691 Mobile & Wireless Sensing] <your subject>*.
- Response time for emails is typically within 24 hours on weekdays.

Reading Materials

The class does not require a mandatory textbook. However, if you wish to deepen your understanding of mobile and wireless sensing, I recommend reading papers from conferences such as MobiCom, MobiSys, SenSys, IPSN, PerCom, IMWUT, and others.

Grading

- Paper Review: 40%
- Research Project: 60%
 - 10% for project proposal presentation & report
 - 10% for project milestone presentation & report
 - 20% for final presentation & demonstration
 - 20% for final report

Paper Reviews

- You will review and discuss 8 assigned research papers. Those papers can be downloaded from Blackboard. Please follow these guidelines when writing your review:
 - Summary: Briefly summarize the key points of the paper.
 - Strength: Identify and discuss the strengths of the paper.
 - Weakness: Critique the paper by addressing any limitations or areas that could have been improved.
- **Deadlines:** You are expected to read papers and submit reviews via *Blackboard* by **11:59 PM** each Wednesday.
- **Late Submission:** No late submissions allowed.

Research Project

- The research project is a core component of the course and will be conducted in groups of 3 or 4 students. You will propose, design, and execute a project that aligns with the course topics. It is acceptable—and often beneficial—to work on a class project that complements your ongoing research. If you're unsure whether your research topic would be appropriate, please discuss it with Dr. Dong Li.
- **Deadlines:** The project milestones and rough timeline are as follows.
 - o Proposal Report (1-2 pages): Due by 11:59 PM on March 3
 - o Milestone Report (3-4 pages): Due by 11:59 PM on March 31
 - o Final Report (7-8 pages): Due by 11:59 PM on May 18
- **Late Submission:** 10% deducted for each late for **all group members**.

CMSC 491 vs. CMSC 691

The lectures and class materials are identical for both the undergraduate-level course (CMSC 491) and the graduate-level course (CMSC 691). However, the primary distinction lies in the scope and expectations of the projects. Graduate students are expected to develop a project centered around an original and unique research hypothesis with the potential for publication. Undergraduate students are also encouraged to propose original and unique research hypotheses but may alternatively choose to work on an idea provided by the instructors.

Course Schedule

The class schedule is tentative and subject to change.

#	Date	Topic	Paper Review Assigned
1	Mon, Jan 26, 2026	No Class - Campus closed	
2	Wed, Jan 28, 2026	No Class - Campus closed	
Module 1: Foundations: Signals, Physics, and Sensing Principles			
3	Mon, Feb 02, 2026	Course Overview & The Vision of Ubiquitous Sensing	
4	Wed, Feb 04, 2026	Physics of Sensing: Waves, Fields, and Human Interaction	Paper 1
5	Mon, Feb 09, 2026	Signals & Systems for Sensing Applications	
Module 2: Wireless Sensing			
6	Wed, Feb 11, 2026	<i>Guest Lecture: Privacy-aware AI-driven Sensing Systems for Health and Wellness (PhD Cand. Bhawana Chhagani)</i>	Paper 2
7	Mon, Feb 16, 2026	Acoustic, Optical, and LiDAR Sensing	
8	Wed, Feb 18, 2026	mmWave, LTE, and WiFi Sensing	Paper 3
9	Mon, Feb 23, 2026	Low-Power & Backscatter Wireless Sensing: LoRa and RFID	
Module 3: Mobile & Wearable Sensing			
10	Wed, Feb 25, 2026	<i>Guest Lecture: Wearable Intelligence for Human Behavior and Health (PhD Cand. Hao Zhou)</i>	
11	Mon, Mar 02, 2026	Mobile Sensing Platforms & Smartphone Sensor Pipelines	
12	Wed, Mar 04, 2026	Project Proposal Presentation and Discussion	Paper 4
13	Mon, Mar 09, 2026	Wearable & On-body Sensors: IMU, PPG, ECG, and Beyond	
Module 4: Earable & Smart Glasses Sensing			
14	Wed, Mar 11, 2026	Beyond the Ear: Earphone-based Sensing for Healthcare	Paper 5
15	Mon, Mar 16, 2026	No Class - Spring Break	
16	Wed, Mar 18, 2026	No Class - Spring Break	
17	Mon, Mar 23, 2026	Smart Glasses Sensing: Vision, Audio, and On-body Interaction	
18	Wed, Mar 25, 2026	<i>Guest Lecture: Earphone-based Sensing (Dr. Tao Chen)</i>	
Module 5: AI-enabled Sensing Systems			
19	Mon, Mar 30, 2026	Machine Learning for Sensing: From Features to Models	
20	Wed, Apr 01, 2026	Project Milestone Presentation and Discussion	Paper 6
21	Mon, Apr 06, 2026	Deep Learning & Foundation Models for Sensing	
22	Wed, Apr 08, 2026	<i>Guest Lecture: LLM-enabled Sensing (Dr. Huatao Xu)</i>	Paper 7
Module 6: Emerging Paradigms & Frontiers in Ubiquitous Sensing			
23	Mon, Apr 13, 2026	<i>Guest Lecture: Microstructure-assisted Sensing (Dr. Nakul Garg)</i>	
24	Wed, Apr 15, 2026	Project Discussion I	Paper 8
25	Mon, Apr 20, 2026	<i>Guest Lecture: Cross-technology Sensing (Dr. Binbin Xie)</i>	
26	Wed, Apr 22, 2026	<i>Guest Lecture: AI-enabled American Sign Language Learning (Dr. Yincheng Jin)</i>	
27	Mon, Apr 27, 2026	<i>Guest Lecture: Multi-modal Sensing (Dr. Ke Sun)</i>	
28	Wed, Apr 29, 2026	Project Discussion II	
Project Presentation and Demonstration			
29	Mon, May 4, 2026	Project Presentation and Demonstration I	
30	Wed, May 6, 2026	Project Presentation and Demonstration II	
31	Mon, May 11, 2026	Project Presentation and Demonstration III	

Class Policies

- Academic Integrity**

Plagiarism or any form of cheating is not tolerated in this class. All submitted work must be your own. If you are suspected of cheating, plagiarism, or other forms of academic dishonesty, your case will be referred to the Undergraduate Academic Conduct Committee or the Graduate Council Grievance Committee, and this may result in lost credit, course failure, suspension, or dismissal from UMBC, depending on the Committee's decision. For more information on what constitutes academic dishonesty, please consult [UMBC Academic Conduct](#). If you are struggling with the class content in any way, it is your responsibility to communicate with the instructor so that he can assist you.

- Generative AI Tools**

The use of generative AI tools (including but not limited to ChatGPT) is not allowed for paper reviews, written reports, or project write-ups in this course. These assignments must reflect the student's own analysis, understanding, and writing. Limited use of generative AI may be permitted for tasks such as code debugging or exploring general technical concepts, provided that AI-generated content is not

directly included in any submitted work. Any misuse of generative AI will be considered a violation of academic integrity. Students should consult the instructor if there is any uncertainty regarding acceptable use.

- **Collaboration Policy**

You are only allowed to work in groups on the research project. When submitting the report, you must clearly indicate the names of all group members and specify each person's contributions to the assignment.

UMBC School Policies

- **Accessibility and Disability Accommodations, Guidance and Resources**

Accommodations for students with disabilities are provided for all students with a qualified disability under the Americans with Disabilities Act (ADA & ADAAA) and Section 504 of the Rehabilitation Act who request and are eligible for accommodations. The Office of Student Disability Services (SDS) is the UMBC department designated to coordinate accommodations that creates equal access for students when barriers to participation exist in University courses, programs, or activities.

If you have a documented disability and need to request academic accommodations in your courses, please refer to the SDS website at sds.umbc.edu for registration information and office procedures.

SDS email: disAbility@umbc.edu

SDS phone: [410-455-2459](tel:410-455-2459)

If you will be using SDS approved accommodations in this class, please contact the instructor to discuss implementation of the accommodations. During remote instruction requirements due to COVID, communication and flexibility will be essential for success.

- **Sexual Assault, Sexual Harassment, and Gender Based Violence and Discrimination**

[UMBC Policy](#) in addition to federal and state law (to include Title IX) prohibits discrimination and harassment on the basis of sex, sexual orientation, and gender identity in University programs and activities. Any student who is impacted by sexual harassment, sexual assault, domestic violence, dating violence, stalking, sexual exploitation, gender discrimination, pregnancy discrimination, gender-based harassment, or related retaliation should contact the University's Title IX Coordinator to make a report and/or access support and resources. The Title IX Coordinator can be reached at titleixcoordinator@umbc.edu or 410-455-1717.

You can access support and resources even if you do not want to take any further action. You will not be forced to file a formal complaint or police report. Please be aware that the University may take action on its own if essential to protect the safety of the community.

If you are interested in making a report, please use the [Online Reporting/Referral Form](#). Please note that, if you report anonymously, the University's ability to respond will be limited.

Notice that Faculty and Teaching Assistants are Responsible Employees with Mandatory Reporting Obligations

All faculty members and teaching assistants are considered Responsible Employees, per UMBC's [Policy on Sexual Misconduct, Sexual Harassment, and Gender Discrimination](#). Faculty and teaching assistants therefore required to report all known information regarding alleged conduct that may be a violation of the Policy to the Title IX Coordinator, even if a student discloses an experience that occurred before attending UMBC and/or an incident that only involves people not affiliated with UMBC. Reports are required regardless of the amount of detail provided and even in instances where support has already been offered or received.

While faculty members want to encourage you to share information related to your life experiences through discussion and written work, students should understand that faculty are required to report past and present sexual harassment, sexual assault, domestic and dating violence, stalking, and gender discrimination that is shared with them to the Title IX Coordinator so that the University can inform students of their [rights, resources, and support](#). While you are encouraged to do so, you are not obligated to respond to outreach conducted as a result of a report to the Title IX Coordinator.

If you need to speak with someone in confidence, who does not have an obligation to report to the Title IX Coordinator, UMBC has a number of [Confidential Resources](#) available to support you:

[Retriever Integrated Health](#) (Main Campus): 410-455-2472; Monday – Friday 8:30 a.m. – 5 p.m.; For After-Hours Support, Call 988.

[Center for Counseling and Well-Being](#) (Shady Grove Campus): 301-738-6273; Monday-Thursday 10:00a.m. – 7:00 p.m. and Friday 10:00 a.m. – 2:00 p.m. (virtual) [Online Appointment Request Form](#).

Pastoral Counseling via [The Gathering Space for Spiritual Well-Being](#): 410-455-6795; i3b@umbc.edu; Monday – Friday 8:00 a.m. – 10:00 p.m.

Other Resources

[Women's Center](#) (open to students of all genders): [410-455-2714](tel:410-455-2714);

womenscenter@umbc.edu;

Monday – Thursday 9:30 a.m. – 5:00 p.m. and Friday 10:00 a.m. – 4 p.m.

[Shady Grove Student Resources](#), [Maryland Resources](#), [National Resources](#).

Child Abuse and Neglect

Please note that Maryland law and [UMBC policy](#) require that faculty report all disclosures or suspicions of child abuse or neglect to the Department of Social Services and/or the police even if the person who experienced the abuse or neglect is now over 18.

- **Pregnant and Parenting Students**

UMBC's [Policy on Sexual Misconduct, Sexual Harassment and Gender Discrimination](#) expressly prohibits all forms of discrimination and harassment on the basis of sex, including pregnancy. Resources for pregnant, parenting and breastfeeding students are available through the University's [Office of Equity and Civil Rights](#). Pregnant and parenting students are encouraged to contact the Title IX Coordinator to discuss plans and ensure ongoing access to their academic program with respect to a leave of absence – returning following leave, or any other accommodation that may be needed related to pregnancy, childbirth, adoption, breastfeeding, and/or the early months of parenting.

In addition, students who are pregnant and have an impairment related to their pregnancy that qualifies as disability under the ADA may be entitled to accommodations through the [Office of Student Disability Services](#).

- **Religious Observances & Accommodations**

UMBC [Policy](#) provides that students should not be penalized because of observances of their religious beliefs, and that students shall be given an opportunity, whenever feasible, to make up within a reasonable time any academic assignment that is missed due to individual participation in religious observances. It is the responsibility of the student to inform the instructor of any intended absences or requested modifications for religious observances in advance, and as early as possible. For questions or guidance regarding religious observances and accommodations, please contact the Office of Equity and Civil Rights at ecr@umbc.edu.

- **Hate, Bias, Discrimination and Harassment**

UMBC values safety, cultural and ethnic diversity, social responsibility, lifelong learning, equity, and civic engagement.

Consistent with these principles, [UMBC Policy](#) prohibits discrimination and harassment in its educational programs and activities or with respect to employment terms and conditions based on race, creed, color, religion, sex, gender, pregnancy, ancestry, age, gender identity or expression, national origin, veterans status, marital status, sexual orientation, physical or mental disability, or genetic information.

Students (and faculty and staff) who experience discrimination, harassment, hate, or bias based upon a protected status or who have such matters reported to them should use the [online reporting/referral form](#) to report discrimination, hate, or bias incidents. You may report incidents that happen to you anonymously. Please note that, if you report anonymously, the University's ability to respond may be limited.